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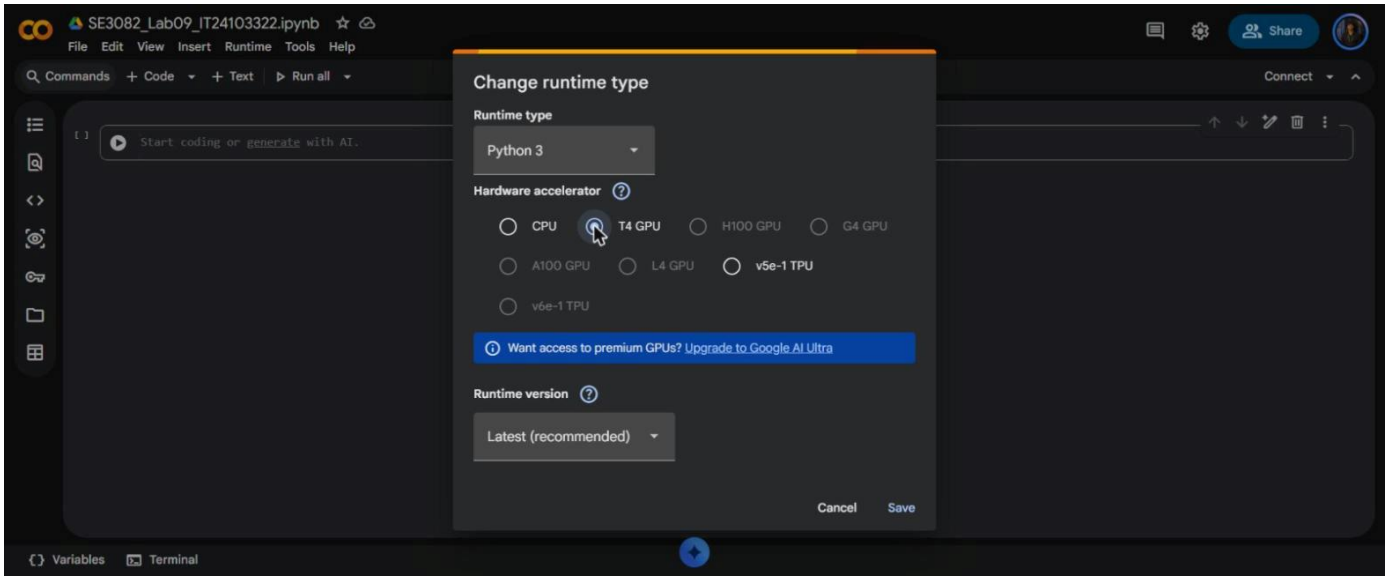
BSc (Hons) in Computer Science
Year 3

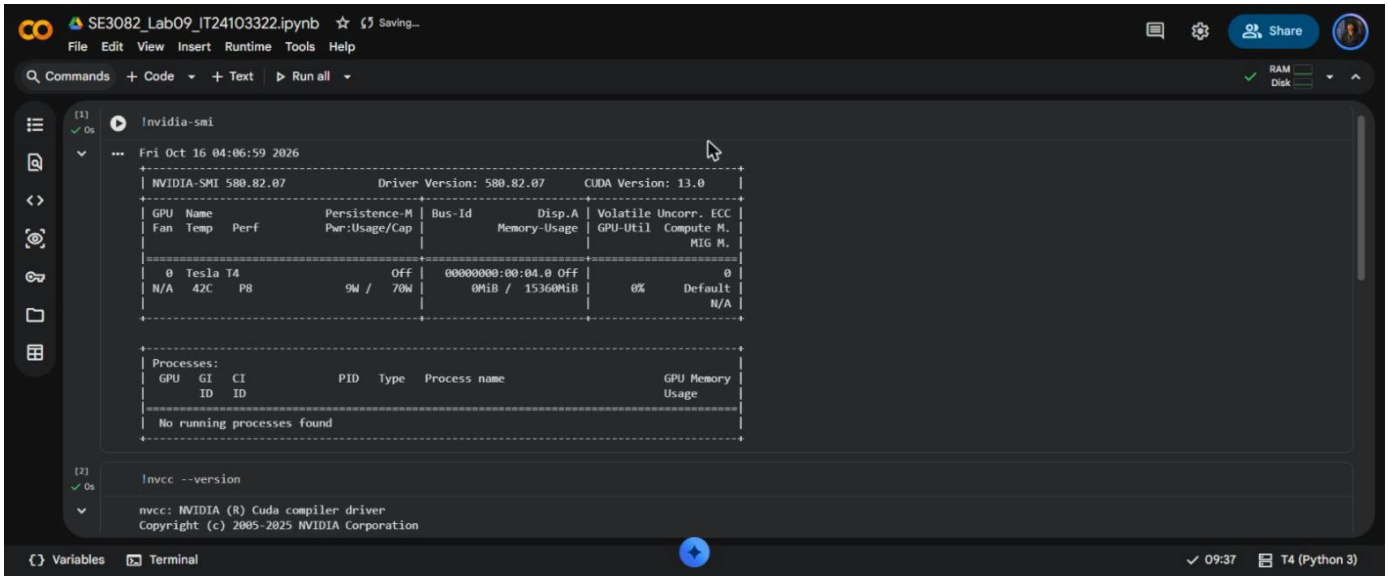
Lab Sheets

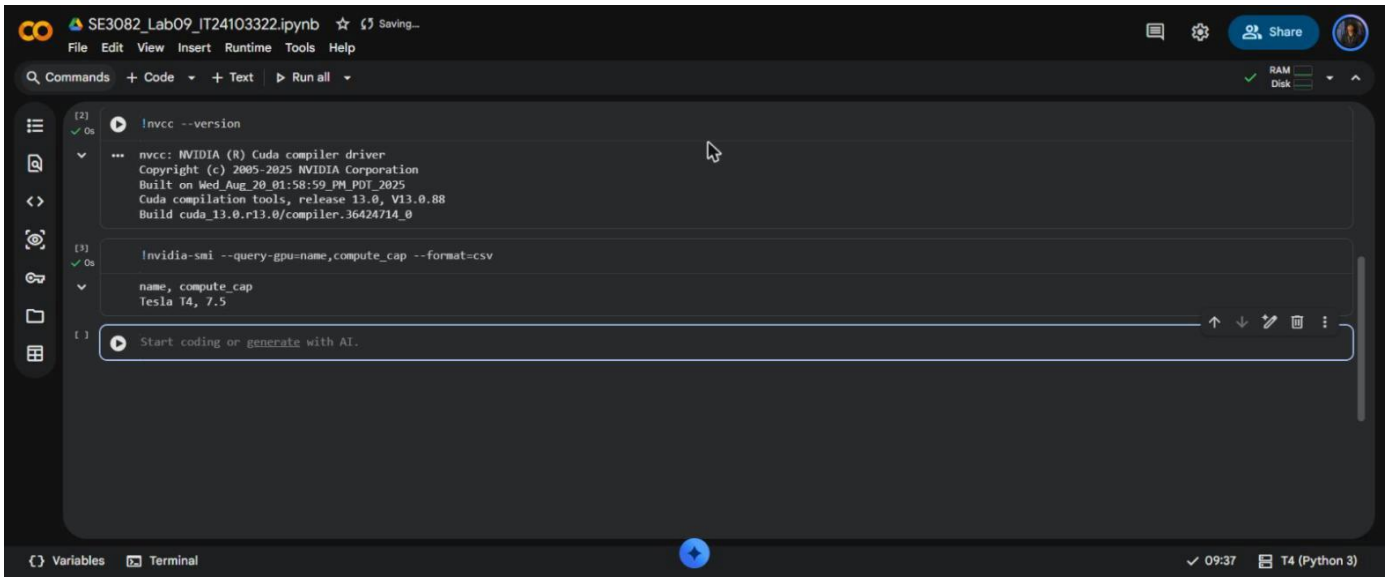
SE3082 – Parallel Computing **Semster 1, 2026**

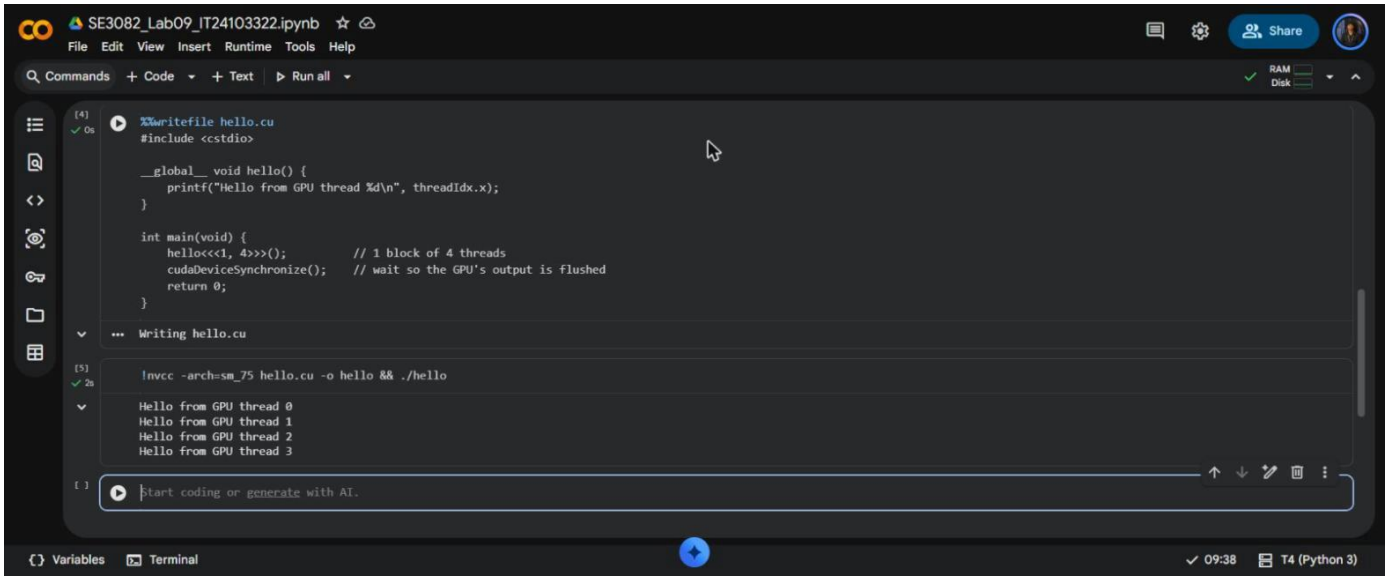
IT24103322

Exercise 1









A new notebook was created in Google Colab and renamed to SE3082_Lab09_IT24103322. Under Runtime → Change runtime type the hardware accelerator was set to T4 GPU and saved.

!nvidia-smi shows the GPU: Tesla T4 with 15360 MiB of memory (driver 580.82.07, CUDA Version 13.0). !nvcc --version shows Cuda compilation tools, release 13.0 (V13.0.88), and the compute_cap query returns Tesla T4, 7.5, so the compiler flag is -arch=sm_75, as used in every compile cell.

hello.cu was written with %%writefile, compiled with nvcc and run. hello<<<1, 4>>>() launches one block of 4 threads, so four lines are printed, from GPU thread 0 to thread 3. cudaDeviceSynchronize() makes the CPU wait until the kernel has finished, so the GPU's printf output is flushed before the program ends. The order of the lines is not fixed, because the four threads run at the same time.